SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Answer Script Evaluation Sheet

Continuous Internal Assessment 1/2

Date: 3.11.2026

Programme	B. Tech	**Slot**	D
Course Code	CSA0709 / CSA0710	**Course Name**	Computer Networks
Duration	1.30 Hrs	**Marks**	50

Q. No.	Max. Marks	Awarded Marks
1.	10	10
2.	10	9
3.	10	9
4.	10	7
5.	10	7
Total	50	42

Course Faculty: Dr. Shankar R.,

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Programme: **B.E / B.Tech**

Course Code: **CSA07**

Course Name: **Computer Networks**

Duration: **1: 30 Hr**

Date : 03.11.2026 Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Explain the layered network architectures (OSI/TCP-IP), the functions of each layer, and the fundamentals of data communication in computer networks.
CO2	Apply IP addressing, subnetting, and routing concepts to design and configure interconnected networks.

Answer All Questions

1. Answer the following questions on the OSI model and Transport layer services.

   (a) Explain the primary functions of the Data Link layer (framing, physical (MAC) addressing, medium access control, error detection using CRC, and hop-to-hopflow control) and of the Transport layer (process-to-process delivery using port numbers, segmentation and reassembly, end-to-end flow control, congestion control, and error recovery through acknowledgements and retransmission). Analyze how the two layers work together to give error-free and reliable data delivery: the Data Link layer detects and discards corrupted frames on each individual link, while the Transport layer provides end-to-end reliability across multiple links and routers by using sequence numbers, ACKs, and retransmission of lost segments. Use a layered diagram to support the explanation? *(6 Marks)*

   (b) Differentiate between connection-oriented service (TCP: handshake before transfer, reliable, ordered, flow and congestion control) and connectionless service (UDP: no setup, best-effort, low overhead) at the Transport layer. Compare them on connection setup, reliability, speed, header size, and state maintained. Give real-world examples: TCP for web browsing (HTTP/HTTPS), email (SMTP), and file transfer (FTP), and UDP for DNS queries, VoIP, video streaming, and online gaming. Present the comparison in a table. *(4 Marks)*

(10) (CO1) (BL2) (BL4)

2. A data packet travels from a source host to a destination host across multiple routers.

   (a) Describe the concept of encapsulation, in which each OSI layer at the source adds its own header (and a trailer at the Data Link layer) to the data received from the layer above, producing the PDUs: data, segment or datagram, packet, frame, and bits. Describe decapsulation, in which each layer at the destination strips its corresponding header and passes the payload upward until the application receives the original data. Explain what happens at intermediate routers (the frame is decapsulated up to the Network layer, the next hop is determined, and a new frame is built). Support the answer with a neat diagram showing the headers at each layer. *(6 Marks)*

   (b) Identify the Network layer as the layer responsible for routing packets across interconnected networks. Explain how it uses logical addressing (IP addresses) to identify the source and destination networks and hosts, how routers use routing tables and the destination IP address to select the best path, and how logical (IP) addresses differ from physical (MAC) addresses, which change at each hop while IP addresses stay the same end to end. Mention fragmentation and the TTL field briefly. *(4 Marks)*

(10) (CO1) (BL2)

3. Answer the following questions on IP addressing and subnetting.

(a) Routers and Layer 3 devices between a source and a destination system forward packets using IP addresses and routing tables. Explain the limitations of classful IP addressing: fixed network and host boundaries (Class A, B, C), wasted addresses (a Class C block of 254 hosts is too small and a Class B block of 65,534 is too large for most organizations), rapid IPv4 address exhaustion, inflexible subnetting, and the growth of routing tables on routers. Explain how Classless Inter-Domain Routing (CIDR) overcomes these problems through variable-length prefixes in slash notation (such as /22), allowing blocks of any suitable size, efficient address allocation, and route aggregation (supernetting) that reduces the size of routing tables and the load on routers. Give a short example of aggregating several contiguous networks into one prefix. *(5 Marks)*

(b) A customer site connects to a service provider's MPLS backbone through its Customer Edge (CE) router, which has the interface address 192.168.25.130 with the subnet mask 255.255.255.192 (/26). The provider edge (PE) router needs the customer's address block in order to associate it with a label-switched path (LSP) or VPN route. Determine, showing the binary working: (i) the Network ID (by ANDing the address with the mask), (ii) the Broadcast Address (by setting all host bits to 1), and (iii) the range of usable host IP addresses (first usable to last usable), along with the block size and the number of hosts in the block. *(5 Marks)*

(10) (CO1)(CO2) (BL2)

4. An organization is assigned the network address 172.16.0.0/16 and must divide it into at least 6 subnets for different departments while maximizing the number of usable hosts per subnet. Show the working at each step.

(a) Determine the minimum number of bits to borrow from the host portion (using $2^n \geq 6$) so that the usable hosts are maximized. Write the resulting subnet mask in both dotted-decimal and CIDR notation.*(3 Marks)*

(b) Calculate the total number of subnets created (2^n) and the number of usable host IP addresses per subnet ($2^h - 2$, where h is the number of remaininghost bits). Comment on the number of unused subnets left over. *(3 Marks)*

(c) For the first two subnets, list the Network ID, first usable IP, last usable IP, and Broadcast address. Show the block size (increment) used to find the second subnet, and present the results in a table. *(4 Marks)*

(10) (CO2) (BL3)

5. Answer the following questions on routing in network simulation tools.

(a) Compare and contrast static routing and dynamic routing in network simulation tools such as Cisco Packet Tracer. Compare them on configuration effort (manual versus automatic), adaptability to topology changes and link failures, convergence, scalability, router CPU and memory use, bandwidth overhead of routing updates, security, and administrative distance. Mention dynamic protocols such as RIP, OSPF, and EIGRP, and explain how a simulator lets you observe their behaviour. Discuss the conditions in which static routing is preferred, such as small networks, stub networks with a single exit path, default routes to an ISP, and situations needing predictability, low overhead, or control. Present the comparison in a table. (6 Marks)

(b) Outline the basic CLI configuration steps on a router to set up a static route directing traffic for the network 192.168.2.0/24 through the next-hop IPaddress 10.0.0.2. Include thesteps enable, configure terminal, ip route 192.168.2.0 255.255.255.0 10.0.0.2, exit, and saving the configuration with copy running-config startup-config. Then show how to verify the route with show ip route (looking for the S code) and test connectivity using ping. (4 Mark

CIA-1

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1) a) Data Link Layer (38)

Data link layer is responsible for a transferring data between two directly connected devices over a physical network.

Functions:

1) Framing: Divides data into frames for transmission

2) Physical addressing: Uses MAC addresses to identify devices on a local network.

3) Error detection:- Detects transmission errors using methods such as CRC.

4) Flow control:- Helps prevent a sender from overwhelming a reciever when supported by the link protocol.

2. Transport Layer:

The transport layer provides communication between applications running on different devices.

* Segmentation: Divides application data into smaller segments.

* Error recovery: TCP detects missing data

b)

Feature	Connection-oriented service.	Connection less service.
Protocol	TCP	UDP
Connected Setup	Required.	Not required.
Reliability.	Supports reliable delivery.	No delivery guarantee.
Acknowledgement	Uses acknowledgements	No built in acknowledgements
Data ordering	Maintains ordered delivery.	No ordering guarantee
Speed and overhead	Generally more overhead	Generally lower overhead.
Example	File transfer, Web browsing using HTTPS, over TCP	Online gaming DNS queries live streaming using UDP

TCP is preferred when reliable, ordered delivery is required, whereas UDP is useful when low latency and low overhead are important

2) a) Encapsulation:

It is a process of adding protocol information to data as it moves down the OSI layers at the source device

OSI layer	process	Data link.
7. Application.	Creates application data	Data.
6 Presentation	Handles translation, encryption or the compression as needed.	Data.
5. Session	Manages communication sessions	Data
4. Transport	Adds transport header, such as TCP or UDP	Segment.
3. Network.	Adds MAC addressing and frame checking information	Packet.
2. Data link	Address source	Frame.
1. Physical	Transmits bits over the medium	Bits

b) The Network layer (Layer 3) of the OSI model is responsible for routing packets from the source to the destination across interconnected networks.

1. Logical addressing: It uses logical addresses (IP addresses) to identify the source and destination networks and hosts, ensuring packets reach the correct destination.

2. Routing: Routers examine the destination IP addresses that generally remain the same from source to destination (end to end). MAC addresses are physical addresses.

3. IP Address VS MAC address: IP addresses are logical addresses that generally remain the same from source to destination (end to end) MAC addresses are physical addresses used for local network delivery and change at each hop as the packet is forwarded through different networks.

4. Fragmentation and TTL: The Network layer performs fragmentation when packets are too large for a networks MTU. The TTL field is decreased by each router to prevent packets from circulating endlessly; when it reaches zero, the packet is discarded

3) a) Classful IP Addressing:

In classful adressing, IPV4 addresses were divided into fixed classes A, B, C, D and E.

class	First Octet Range	Default Mask	Purpose.
A.	1-126	255.00(18)	Large networks
B	128-191	255.255-0.0 (/16)	Medium sized networks.
C	192-223	255.255.0(/24)	Small Networks
D	224-239	Not applicable	Multicast
E	240255	Not applicable	Reversed

CIDR improves IPV4 address utilization and routing efficiency by using flexible network prefix lengths. Smaller organizations can recieve smaller address blocks.

b) Given:

IP address: 192.168.25.130.

Subnet Mask: 255.255.255.192

CIDR notation: /26.

A /26 subnet means that 26 bits are used for the network portion and 6 bits are used for the host portion.

Convert IP address and subnet mask into binary.

IP Address	Binary.
192	11000000
168	10101000
25	00011001
130	10000010.

IP address in binary

11000000.10101000.00011001.10000010.

(i) Network ID

Network ID in decimal

192.168.25.128.

(ii) Broadcast address.

The /26 subnet has 6 host bits set to 1:

192.168.25.191

First usable IP: 192.168.25.129.

Last usable IP = Broadcast address - 1
= 192.168.25.190

1. Block size and Number of Hosts

Block size = 64
No. of host bits: 6
Total IP address = $2^6 = 64$
usable host addresses: $2^6 - 2 = 62$

4) a) Given;
Network address: 172.16.0.0 /16.
Required subnets: At least 6.

Step 1:

$2^n \geq$ Required subnets.

For 6 subnets:

$2^2 = 4$
$2^3 = 8$.
∴ 3 bits must be borrowed

Step 2:

prefix = /16
Borrowed = 3
New prefix = 16+3 = 19.

∴ new CIDR prefix is /19 255.255.244.0.

∴ CIDR notation: /19
Dotted decimal subnet mask: 255.255.244.0

b) Formula:

$2^n = 2^3 = 8$

Total Subnets created = 8.

Remaining Host bits $h = 32 - 19 = 13$

No of usable Host/addresses

$2^n - 2$

Unused Subnets $= 2^{10} - 2 = 2190$

$8 - 6 = 2$.

c) Subnet mask:

255.255.224.0

Block size $= 256 - 224 = 32$

Network IP:

172.16.0.0/19.

* First usable IP 172.16.0.1
* Broadcast address: 172.16.31.255
* Last usable IP: 172.16.31.254.

Second subnet = 32.

Network ID:

172.16.32.0/19

First usable IP: 172.16.32.1

Broadcast address: 172.16.63.255.

Last usable IP: 172.16.63.254

5) a) Static Routing:- Routes are manually configured by the network administrator. They do not automatically change when the network topology changes.

Dynamic Routing: Routers automatically learn & update routes using routing protocols such as RIP, OSPF and EIGRP.

Criteria	Static Routing	Dynamic Routing
Configuration effort	Routes are manually configured by the administrator	Routes are learned automatically using routing protocols.
Topology changes	Does not automatically adapt to network changes	Automatically adapts to topology changes

b) Static Route configuration:

Router > enable.

Router # configure terminal

Router (config) # ip route 192.168.2.0. 255.255.255.0.10.0.0.2.

Router (config) #end

Router #show IP route